Kaustubh Hemant Sonawane

Bristol, UK | ksstubh.sonawane@gmail.com | +44 7388229743 | +91 8591441019 | www.github.com/Arishtanemi4 |
www.linkedin.com/in/kaustubhson4 | https://kaustubh-sonawane-portfolio.vercel.app

UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Software engineer with 3 years of industry experience building and supporting Python systems end to end: **FastAPI microservices in Docker**, **PostgreSQL** data platforms and internal **GenAI** applications on **AWS** with **Jenkins and GitLab CI/CD**. Built **reusable Python components and APIs** - SQL-backed tool functions, ingestion and embedding pipelines and a ReactJS frontend - used across internal applications, delivering directly with Client Service, Compliance and Investment stakeholders; acted as **de facto team lead** for up to 4 interns and was twice recognised with **Best Team Project (2024, 2025)**. Holds an **MSc in Data Science** (University of Bristol, awaiting result), including a multi-architecture deep learning study and an algorithmic dissertation with AstraZeneca. Seeking to build the Python libraries and platform tooling that help machine learning teams train and deploy models at scale, and to work on embodied AI for autonomy.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Python Engineering, APIs & Reusable Components

- Designed and built internal applications end to end for Compliance, Client Service and Investment: a **ReactJS** frontend and **Python FastAPI microservices** (**Docker**, uvicorn) with the GenAI and RAG logic behind them - initially the only engineer, later coordinating a frontend developer and **up to 4 interns**.
- Built **reusable Python components** other applications depend on: **SQL-backed tool functions** that render fund-comparison PDFs on request, a document **ingestion and embedding pipeline**, and an **LDAP authentication function** powering **role-based access control** for three user tiers.
- Engineered for performance where it mattered: wrote **ExpressJS APIs** in development because the Python calculation endpoints were the bottleneck, ran document processing as a **scheduled AWS job** (EventBridge, Lambda, SageMaker) and **versioned every prompt change in Git**.

GenAI & Retrieval Engineering

- Developed internal **GenAI applications** with **LlamaIndex** and **Azure OpenAI** (GPT-3.5 & Llama-3) answering **department-specific questions** over internal documents.
- Designed and maintained a **vector database** on **pgvector** with **Azure OpenAI embeddings**, held **per department** so each team's documents and embeddings stayed siloed and access was enforced.
- Made **cost-aware model choices**: self-hosted **Llama-3 on a GPU EC2 instance** for the largest department, keeping the stronger **Azure OpenAI** outputs for decision makers, with the ability to switch models.

Data Pipelines & Data Engineering

- Designed the **Normalized SQL Schema** for the internal analytics platform (**PostgreSQL**, with **pgvector** in a separate schema) and developed **Optimized SQL queries**, **Functions** and **Procedures** for efficient preprocessing and insertion.

- Wrote the **Python ingestion scripts** calling the enterprise-licensed **Morningstar API** for **every fund in the Indian mutual fund market (2023-2025)**, refreshed fortnightly and monthly, and merged it with legacy data held in analysts' Excel files.
- Built a **Python automation job** on **AWS SageMaker** (EventBridge 9 AM trigger, Lambda) that parses **over 300 daily** rating-agency emails into structured data, fuzzy-matches tracked companies and emails analysts a daily summary, **saving 60 analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** for image emails.

Cloud Infrastructure, CI/CD & Security

- Designed the **infrastructure architecture** for the internal analytics platform and GenAI applications on **AWS** (**EC2, S3, ECR, Lambda**), with separate frontend and backend repositories and Docker containers for dev and prod.
- Built **one-click deployment** in **Jenkins and GitLab**, a **CI/CD pipeline** that ships the chosen branch to dev or prod, and provisioned new **EC2** instances using **CloudFormation** templates from the security team.
- Worked with the security team on port whitelisting and safeguards, restricted access to **company private IPs** with **AWS Security Groups**, added **RBAC** against **LDAP** and deployed an **Nginx reverse proxy**, one of the fixes that took the applications from failing **VAPT** to passing.

PROJECTS

- *University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca*: **CellLineSelector** (Team project, co-built. A preprocessing pipeline turning 17 raw inputs into 14 clean joined tables with integrity checks, a per-gene scoring algorithm and a RAG layer, built as **five FastAPI services** with a **React and TypeScript** frontend; ranks around **2,100 cancer cell lines** on target inclusion and exclusion genes, validated with AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **B.Tech Final Year Project**: Udacity **self-driving car simulator**. (A self-driving car trained using a **Convolutional Neural Network (CNN)** model - my first work on autonomy.)
- *University of Bristol MSc Applied Data Science Group Project*: **VisionCaptioner** (Team project. Multi-phase architecture study comparing **CNN, ResNet-18 and ViT encoders** with symbolic, embedding and LSTM/attention decoders across three synthetic datasets of 10,000 images and five seeds, benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **Wingman**: AI dating-coach web app (**ReactJS**, **Python FastAPI**) with **108 LLM personas**. Rate limits shaped the architecture: LLM and vision calls **serialised behind a semaphore**, results **streamed over server-sent events**, responses **cached by input hash**, and LLM output validated with **Pydantic** and repaired rather than trusted. GitHub: https://github.com/Arishtanemi4/wingman.git

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ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git

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SKILLS

- **Tools**: Git, Docker, Jenkins, GitLab (CI/CD), AWS (EC2, S3, ECR, Lambda, EventBridge, SageMaker), Azure OpenAI, Nginx, Linux, Power BI, Claude Code.
- **Languages**: Python (FastAPI, Pydantic, pandas, scikit-learn), SQL, NodeJS (Express, ReactJS), R (dplyr, tidyverse, ggplot2), Shell; concurrent and asynchronous Python (async services, rate-limited parallel calls, SSE streaming).
- **Databases**: PostgreSQL (pgvector, SQL functions and procedures), SQLite

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